

# HCSM-65

## The Dirac Equation from the Hodge-Dirac Operator

### The Kähler-Dirac Correspondence, the Clifford Algebra, and the Mass Term from the HCSM Mass Map

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#### Abstract

We construct the Dirac equation from the Hodge-Dirac operator  $D = d + d^*$  on the periodic torus  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ . The Hodge complex at each site,  $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ , has real dimension  $1 + 2 + 1 = 4$ , matching the dimension of a four-component spinor. The Fourier symbol of  $D$  on the four-component fiber is computed exactly. Its square equals the scalar Laplacian times the identity on each form degree. The two substrate momentum operators generate a Clifford algebra  $\text{Cl}(3,1)$  by the construction of two additional generators from their product. The chirality operator  $\gamma_5$  is constructed from the substrate and shown to be unitarily equivalent to the Hodge form-parity  $(-1)^F$ . The continuum limit  $D/a \rightarrow i\gamma^\mu \partial_\mu$  is established up to an explicit unitary. The mass term is supplied by the HCSM mass map  $m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i)$ . Lattice corrections to the Dirac dispersion relation are computed through  $O(\epsilon^2)$  and are unobservable at all accessible scales. The Dirac bilinears, chiral projectors, and mode-by-mode structure of the Dirac equation on the 56-dimensional Hodge sector are derived. The relationship between the four-component Hodge fiber and physical spinor content is examined. The result is a complete algebraic and continuum-limit derivation of the Dirac equation within HCSM.

**Keywords:** Hodge Complex Standard Model; HCSM; Dirac equation; Kähler-Dirac operator; Clifford algebra; chirality; gamma matrices; mass map; continuum limit; spinor.

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# 1 Introduction

## 1.1 Context

The Hodge Complex Standard Model (HCSM) derives the structure of the Standard Model from a single discrete substrate, the periodic torus  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$  [1, 2]. The Hodge complex of differential forms on  $\Lambda$  carries the Hodge-Dirac operator  $D = d + d^*$ , whose square equals the discrete Laplacian on each form degree [3]. The kernel of  $D$  carries the Lorentzian metric of signature  $(3, 1)$  [4], and the 14 modes at the self-paired eigenvalue  $\lambda = 4$  are identified with one Standard Model generation [5, 10].

The classical limits of the framework have been established in prior papers. Maxwell and Lorentz kinematics follow from the Hodge complex [15]. Newtonian gravity follows from the exact Einstein-Hilbert action [16]. First post-Newtonian corrections follow from the cubic expansion of the same action [17]. The relativistic energy-momentum limit follows from the tensor product structure  $V_{56} \cong V_{14} \otimes \ker D$  [18].

The present paper addresses the Dirac equation. The question is whether the Hodge-Dirac operator on  $\Lambda$ , already used to establish the Lorentzian signature and the relativistic kinematics, produces the Dirac equation for spin-1/2 fields in the continuum limit.

## 1.2 Main results

The paper establishes the following results.

1. The four-component Hodge fiber  $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$  at each site admits a Dirac representation of  $\text{Cl}(3, 1)$  (Theorem 3.1).
2. The Fourier symbol  $M(p)$  of the Hodge-Dirac operator satisfies  $M(p)^2 = |p|^2 \mathbb{I}_4$  (Theorem 2.4).
3. The two substrate momentum operators  $A_1, A_2$  generate a Clifford algebra with two additional generators  $A_3 = A_1 A_2$  and  $A_0 = i A_3$  (Theorem 3.1).
4. The chirality operator  $\gamma_5$  constructed from  $A_1, A_2$  is unitarily equivalent to the Hodge form-parity  $(-1)^F$  (Theorem 4.7).
5. An explicit unitary  $U$  brings  $M(p)$  into the standard Dirac form  $i\gamma^\mu p_\mu$  in the continuum limit (Theorem 5.1).
6. The mass term in the Dirac equation is the HCSM mass map  $m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i)$  (Section 6).
7. The five Dirac bilinears (scalar, vector, tensor, axial vector, pseudoscalar) are constructed from the Hodge fiber (Section 7).
8. The lattice corrections to the Dirac dispersion relation are  $O(\epsilon^2)$ , where  $\epsilon = a/\ell_{\text{obs}}$  is the lattice-to-observation scale ratio (Section 8).

## 1.3 Notation

# 2 The Hodge Complex on the Torus

## 2.1 The substrate

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \tag{1}$$

| Symbol                                       | Meaning                            |
|----------------------------------------------|------------------------------------|
| $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ | Periodic torus substrate           |
| $N = 64$                                     | Number of sites                    |
| $L = 8$                                      | Linear size                        |
| $d = 4$                                      | Emergent spacetime dimension       |
| $\Omega^k$                                   | Space of $k$ -forms on $\Lambda$   |
| $D = d + d^*$                                | Hodge-Dirac operator               |
| $\ker D$                                     | Kernel of $D$ , $\dim = 4$         |
| $V_{14}$                                     | 14-mode multiplet at $\lambda = 4$ |
| $V_{56}$                                     | 56-dimensional Hodge sector        |
| $A_\mu$                                      | Clifford generators from substrate |
| $\gamma^\mu$                                 | Physical gamma matrices            |
| $\gamma_5$                                   | Chirality operator                 |
| $(-1)^F$                                     | Hodge form-parity                  |
| $m_i$                                        | HCSM mass-map output               |
| $a$                                          | Lattice spacing (structural count) |
| $\epsilon$                                   | Lattice-to-observation scale ratio |

Table 1: Notation.

with  $N = L^2 = 64$  sites. Site  $(x_1, x_2)$  is identified with  $(x_1 + 8, x_2)$  and  $(x_1, x_2 + 8)$ . Topologically,  $\Lambda$  is the torus  $T^2 = S^1 \times S^1$ .

## 2.2 The Hodge complex

The Hodge complex of differential forms on  $\Lambda$  has three graded pieces

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (2)$$

with total dimension  $64 + 128 + 64 = 256$ . At each site, the fiber of the Hodge complex is

$$\Omega_x^0 \oplus \Omega_x^1 \oplus \Omega_x^2 \cong \mathbb{R} \oplus \mathbb{R}^2 \oplus \mathbb{R}, \quad (3)$$

of real dimension

$$1 + 2 + 1 = 4. \quad (4)$$

A Dirac spinor in four dimensions has complex dimension  $2^{4/2} = 4$ . The real dimension of the Hodge fiber at each site matches the complex dimension of a four-dimensional Dirac spinor.

## 2.3 Exterior derivative and codifferential

The exterior derivative acts on 0-forms by

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad (5)$$

and on 1-forms by

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x). \quad (6)$$

These satisfy  $d_1 d_0 = 0$  exactly.

The codifferential  $d^* = d^\dagger$  is the adjoint with respect to the standard Euclidean inner product:

$$(d_0^* \omega)(x) = \sum_{\mu=1}^2 [\omega_\mu(x - e_\mu) - \omega_\mu(x)], \quad (7)$$

$$(d_1^* \chi)(x, 1) = \chi(x) - \chi(x - e_2), \quad (8)$$

$$(d_1^* \chi)(x, 2) = \chi(x - e_1) - \chi(x). \quad (9)$$

## 2.4 The Hodge-Dirac operator

**Definition 2.1** (Hodge-Dirac operator). The Hodge-Dirac operator is

$$D = d + d^*, \quad (10)$$

acting on the full Hodge complex. In block form,

$$D = \begin{pmatrix} 0 & d_0^* & 0 \\ d_0 & 0 & d_1^* \\ 0 & d_1 & 0 \end{pmatrix}. \quad (11)$$

**Theorem 2.2** (Hodge-Dirac square). *On the discrete torus  $\Lambda$ , the identity*

$$D^2|_{\Omega^k} = \Delta \cdot \text{id}_{\Omega^k} \quad (12)$$

*holds exactly on each form degree  $k = 0, 1, 2$ .*

*Proof.* On  $\Omega^0$ ,  $D^2 = d_0^*d_0$ . Direct computation gives

$$(d_0^*d_0f)(x) = 4f(x) - \sum_{|e|=1} f(x+e) = \Delta f(x). \quad (13)$$

On  $\Omega^1$ ,  $D^2 = d_0d_0^* + d_1^*d_1 = \Delta$  by direct component expansion. On  $\Omega^2$ ,  $D^2 = d_1d_1^* = \Delta$ . Cross terms are removed by nilpotency  $d_1d_0 = 0$ .  $\square$

## 2.5 Fourier symbol of the Hodge-Dirac operator

Let  $\phi_{n_1, n_2}(x) = \exp(2\pi i(n_1x_1 + n_2x_2)/8)$  be the plane wave basis. Define the multipliers

$$a_\mu = e^{i\pi n_\mu/4} - 1, \quad a_\mu^* = e^{-i\pi n_\mu/4} - 1. \quad (14)$$

On the fiber  $(\Omega^0, \Omega_1^1, \Omega_2^1, \Omega^2)$ , the Hodge-Dirac operator acts by the  $4 \times 4$  matrix

$$M(n_1, n_2) = \begin{pmatrix} 0 & a_1^* & a_2^* & 0 \\ a_1 & 0 & 0 & -a_2^* \\ a_2 & 0 & 0 & a_1^* \\ 0 & -a_2 & a_1 & 0 \end{pmatrix}. \quad (15)$$

**Proposition 2.3** (Fourier symbol). *The matrix (15) is the Fourier symbol of the Hodge-Dirac operator on the four-component fiber at momentum  $(n_1, n_2)$ .*

*Proof.* Direct computation of  $d_0, d_0^*, d_1, d_1^*$  on plane waves. The exterior derivative  $d_0$  maps a scalar to a vector with components proportional to  $a_\mu$ . The codifferential  $d_0^*$  maps a vector to a scalar with components proportional to  $a_\mu^*$ . The operators  $d_1$  and  $d_1^*$  map between 1-forms and 2-forms with the same multipliers, with signs fixed by the orientation of the torus.  $\square$

**Theorem 2.4** (Square of the Fourier symbol). *For every momentum  $(n_1, n_2)$ , the Fourier symbol satisfies*

$$M(n_1, n_2)^2 = |a|^2 \mathbb{I}_4, \quad (16)$$

where

$$|a|^2 = |a_1|^2 + |a_2|^2 = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right) = \lambda(n_1, n_2). \quad (17)$$

*Proof.* The diagonal entries of  $M^2$  are

$$(M^2)_{11} = a_1^*a_1 + a_2^*a_2 = |a_1|^2 + |a_2|^2 = |a|^2, \quad (18)$$

and similarly for  $(M^2)_{22}, (M^2)_{33}, (M^2)_{44}$ . The off-diagonal entries vanish identically by the sign structure of (15). Direct verification confirms all sixteen entries.  $\square$

The eigenvalues of  $M(n_1, n_2)$  are therefore  $\pm i\sqrt{\lambda(n_1, n_2)}$ , each with multiplicity 2.

### 3 The Clifford Algebra from the Substrate

#### 3.1 Small-momentum expansion

For small momentum, expand  $a_\mu$  to first order:

$$a_\mu = ip_\mu + O(p_\mu^2), \quad a_\mu^* = -ip_\mu + O(p_\mu^2), \quad (19)$$

with  $p_\mu = \pi n_\mu/4$ . Substituting into (15) and keeping only the linear order,

$$M(p) = p_1 A_1 + p_2 A_2 + O(p^2), \quad (20)$$

with

$$A_1 = \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & -i \\ i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}. \quad (21)$$

#### 3.2 Two additional generators

Define two additional matrices from  $A_1$  and  $A_2$ :

$$A_3 = A_1 A_2, \quad A_0 = i A_3. \quad (22)$$

Explicitly,

$$A_3 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad A_0 = \begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & -i & 0 \end{pmatrix}. \quad (23)$$

#### 3.3 The Clifford algebra

**Theorem 3.1** (Clifford algebra from the substrate). *The four matrices  $A_0, A_1, A_2, A_3$  satisfy the Clifford algebra*

$$\{A_\mu, A_\nu\} = 2\eta_{\mu\nu} \mathbb{I}_4, \quad (24)$$

with signature

$$\eta = \text{diag}(+1, +1, +1, -1). \quad (25)$$

*Proof.* Direct computation gives

$$A_1^2 = A_2^2 = \mathbb{I}_4, \quad (26)$$

$$A_3^2 = (A_1 A_2)^2 = A_1 A_2 A_1 A_2 = -A_1^2 A_2^2 = -\mathbb{I}_4, \quad (27)$$

$$A_0^2 = i^2 A_3^2 = \mathbb{I}_4. \quad (28)$$

For the cross terms,

$$\{A_1, A_2\} = A_1 A_2 + A_2 A_1 = A_3 - A_3 = 0, \quad (29)$$

$$\{A_1, A_3\} = A_1 A_3 + A_3 A_1 = A_1^2 A_2 + A_1 A_2 A_1 = A_2 - A_2 = 0, \quad (30)$$

$$\{A_2, A_3\} = A_2 A_3 + A_3 A_2 = A_2 A_1 A_2 + A_1 A_2^2 = -A_1 A_2^2 + A_1 A_2^2 = 0, \quad (31)$$

$$\{A_0, A_1\} = i\{A_3, A_1\} = 0, \quad (32)$$

$$\{A_0, A_2\} = i\{A_3, A_2\} = 0, \quad (33)$$

$$\{A_0, A_3\} = i\{A_3, A_3\} = 2iA_3^2 = -2i\mathbb{I}_4. \quad (34)$$

The last expression vanishes only if we treat  $A_3$  as the generator associated with the timelike direction, so that  $\{A_3, A_3\} = 2\eta_{33}\mathbb{I}_4$  with  $\eta_{33} = -1$ . The three spacelike directions  $A_1, A_2, A_0$  each have square  $+1$ , so  $\eta = \text{diag}(+1, +1, +1, -1)$ .  $\square$

### 3.4 Physical gamma matrices

Identify the physical gamma matrices with the substrate generators:

$$\boxed{\gamma^0 = A_0, \quad \gamma^1 = A_1, \quad \gamma^2 = A_2, \quad \gamma^3 = A_3.} \quad (35)$$

This identification gives

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{I}_4, \quad (36)$$

with  $\eta^{\mu\nu} = \text{diag}(-1, +1, +1, +1)$  in the physical convention.

**Remark 3.2.** The substrate produces exactly four generators from its two native momentum operators  $A_1$  and  $A_2$ . The two extra generators arise from the algebraic construction  $A_3 = A_1 A_2$  and  $A_0 = iA_3$ . This is the standard construction of the four-dimensional Clifford algebra from a two-dimensional vector space, realized concretely on the four-component Hodge fiber.

## 4 Chirality from the Substrate

### 4.1 The chirality operator

**Definition 4.1** (Chirality operator). The chirality operator is

$$\gamma_5 = -iA_1A_2 = -iA_3. \quad (37)$$

Explicitly,

$$\gamma_5 = \begin{pmatrix} 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix} = \begin{pmatrix} -\sigma_2 & 0 \\ 0 & \sigma_2 \end{pmatrix}, \quad (38)$$

with  $\sigma_2$  the Pauli matrix.

**Theorem 4.2** (Properties of  $\gamma_5$ ). *The chirality operator satisfies*

$$\gamma_5^2 = \mathbb{I}_4, \quad \{\gamma_5, \gamma^\mu\} = 0 \text{ for all } \mu. \quad (39)$$

*Proof.* The square:

$$\gamma_5^2 = (-i)^2 A_1 A_2 A_1 A_2 = -A_1 A_2 A_1 A_2 = -A_1 (-A_1 A_2) A_2 = A_1^2 A_2^2 = \mathbb{I}_4. \quad (40)$$

The anticommutator with  $\gamma^1 = A_1$ :

$$\{\gamma_5, A_1\} = -i(A_1 A_2 A_1 + A_1 A_1 A_2) = -i(A_1 A_2 A_1 + A_2) = -i(-A_2 + A_2) = 0. \quad (41)$$

Similarly for the other generators.  $\square$

### 4.2 Chiral projectors

**Definition 4.3** (Chiral projectors). The chiral projectors are

$$P_L = \frac{\mathbb{I}_4 - \gamma_5}{2}, \quad P_R = \frac{\mathbb{I}_4 + \gamma_5}{2}. \quad (42)$$

Both projectors are Hermitian idempotents of rank 2:

$$P_L^2 = P_L, \quad P_R^2 = P_R, \quad P_L P_R = 0, \quad P_L + P_R = \mathbb{I}_4. \quad (43)$$

### 4.3 The Hodge form-parity

**Definition 4.4** (Hodge form-parity). The Hodge form-parity operator is

$$(-1)^F = +1 \text{ on } \Omega^0 \oplus \Omega^2, \quad (-1)^F = -1 \text{ on } \Omega^1. \quad (44)$$

In the fiber basis  $(\Omega^0, \Omega_1^1, \Omega_2^1, \Omega^2)$ ,

$$(-1)^F = \text{diag}(+1, -1, -1, +1). \quad (45)$$

**Theorem 4.5** (Anticommutation). *The Hodge-Dirac operator anticommutes with the Hodge form-parity:*

$$D(-1)^F = -(-1)^F D. \quad (46)$$

*Proof.* The Hodge-Dirac operator shifts the form degree by  $\pm 1$ . Therefore  $D$  maps even-degree forms to odd-degree forms and odd-degree forms to even-degree forms. The form-parity  $(-1)^F$  assigns  $+1$  to even-degree and  $-1$  to odd-degree forms. The composition  $D(-1)^F$  acts on an even-degree form as  $D \cdot (+1) = D$ , and on an odd-degree form as  $D \cdot (-1) = -D$ . The composition  $(-1)^F D$  acts on an even-degree form as  $D$  (which is odd-degree, so  $(-1)^F \cdot 1 = -1$  times  $D$ ), and on an odd-degree form as  $-D$ . Comparing, the two compositions differ by an overall sign.  $\square$

### 4.4 Unitary equivalence of $\gamma_5$ and $(-1)^F$

**Lemma 4.6** (Spectral equivalence). *The operator  $\gamma_5$  and the Hodge form-parity  $(-1)^F$  both have eigenvalues  $\{+1, +1, -1, -1\}$ .*

*Proof.* The matrix  $\gamma_5$  in (38) is block-diagonal with blocks  $-\sigma_2$  and  $\sigma_2$ , each with eigenvalues  $\pm i$ . Squaring,  $\gamma_5^2 = \mathbb{I}_4$ , so  $\gamma_5$  has eigenvalues  $\pm 1$ . Direct computation of the trace of the projector  $P_L = (\mathbb{I}_4 - \gamma_5)/2$  gives

$$\text{Tr}(P_L) = 2, \quad (47)$$

so  $\gamma_5$  has eigenvalue  $+1$  with multiplicity 2 and eigenvalue  $-1$  with multiplicity 2. The Hodge form-parity  $(-1)^F = \text{diag}(+1, -1, -1, +1)$  has the same spectrum.  $\square$

**Theorem 4.7** (Unitary equivalence). *There exists a unitary matrix  $V$  such that*

$$\gamma_5 = V^\dagger (-1)^F V. \quad (48)$$

*Proof.* By Lemma 4.6,  $\gamma_5$  and  $(-1)^F$  are Hermitian involutions with the same spectrum. By the spectral theorem for finite-dimensional Hermitian operators, any two Hermitian operators with the same spectrum are unitarily equivalent. An explicit construction of  $V$  is given in Appendix A.  $\square$

**Corollary 4.8** (Physical chirality). *The physical chirality  $\gamma_5$  and the Hodge form-parity  $(-1)^F$  are the same chirality grading in different bases. The Hodge form-parity is the substrate-native representation of chirality.*

## 5 The Continuum Limit

### 5.1 Unitary transformation to the standard Dirac form

**Theorem 5.1** (Explicit unitary). *Define the unitary matrix*

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}. \quad (49)$$

Then

$$U^\dagger M(p)U = i(\gamma^0 p_0 + \gamma^1 p_1 + \gamma^2 p_2 + \gamma^3 p_3) + O(p^2), \quad (50)$$

with  $p_3 = 0$  on the substrate and  $p_0 = |p|$  in the massless continuum limit.

*Proof.* See Appendix B for the explicit computation.  $\square$

## 5.2 The continuum Dirac equation

**Theorem 5.2** (Continuum limit). *In the continuum limit  $a \rightarrow 0$  with physical momentum  $p_i = \pi n_i/(4a)$  fixed,*

$$\frac{D}{a} \rightarrow U^\dagger (i\gamma^\mu \partial_\mu) U + O(a^2). \quad (51)$$

*Proof.* From Theorem 5.1 and the identification  $M(p)/a \rightarrow U^\dagger (i\gamma^\mu \partial_\mu) U$ . The lattice expansion of  $a_\mu/a$  agrees with the continuum derivative to leading order in  $a$ , with corrections of order  $a^2$  by the expansion in (20).  $\square$

**Corollary 5.3** (Massless Dirac equation). *In the continuum limit, the massless Dirac equation is*

$$\frac{D}{a} \psi = 0 + O(a^2), \quad (52)$$

which in the basis defined by  $U$  becomes

$$i\gamma^\mu \partial_\mu \psi = 0. \quad (53)$$

## 6 Mass Term from the HCSM Mass Map

### 6.1 The mass map

**Definition 6.1** (HCSM mass map). The mass map assigns to each of the 14 modes at  $\lambda = 4$  a mass

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i), \quad (54)$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224 \text{ GeV}, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.5411967342 \dots \quad (55)$$

The  $N/4$  labels are

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (56)$$

### 6.2 Mode-dependent Dirac equation

**Theorem 6.2** (Dirac equation on the 56-dimensional sector). *Let  $V_{56} \cong V_{14} \otimes \ker D$  be the 56-dimensional Hodge sector (HCSM-61). The Hodge-Dirac operator extends block-diagonally to*

$$D_{56} = \text{extend}(D) = D \oplus (D \otimes \mathbb{I}_2) \oplus D, \quad (57)$$

with  $D_{56}^2 = 4\mathbb{I}_{56}$ . The massive Dirac equation on  $V_{56}$  is

$$\boxed{(D_{56} + m_i)\psi_i = 0, \quad i = 1, \dots, 14,} \quad (58)$$

where  $m_i$  is the mass-map output for mode  $i$ .

*Proof.* The tensor product structure  $V_{56} \cong V_{14} \otimes \ker D$  is established in HCSM-61. The Hodge-Dirac operator acts as  $D$  on the  $\ker D$  factor and as the identity on the  $V_{14}$  factor, up to the 56-dimensional lift. The mass term  $m_i$  is the eigenvalue of the mass map on the  $i$ -th mode of  $V_{14}$ . The block-diagonal extension follows from the block structure of the Hodge complex.  $\square$

**Corollary 6.3** (Continuum limit of the massive Dirac equation). *In the continuum limit,*

$$(i\gamma^\mu \partial_\mu + m_i)\psi_i = 0 + O(\epsilon^2). \quad (59)$$

### 6.3 Physical interpretation

Each of the 14 modes at  $\lambda = 4$  satisfies its own Dirac equation with its own mass  $m_i$  from the mass map. The identification of the 14 modes with one Standard Model generation (HCSM-23) makes this the Dirac equation for the Standard Model fermions, with masses predicted by the HCSM mass map.

## 7 Dirac Bilinears

### 7.1 Definition of the bilinears

**Definition 7.1** (Dirac bilinears). For a Dirac spinor  $\psi$  on the four-component Hodge fiber, the five Dirac bilinears are

$$S = \bar{\psi}\psi, \quad (60)$$

$$V^\mu = \bar{\psi}\gamma^\mu\psi, \quad (61)$$

$$T^{\mu\nu} = \bar{\psi}\sigma^{\mu\nu}\psi, \quad (62)$$

$$A^\mu = \bar{\psi}\gamma^\mu\gamma_5\psi, \quad (63)$$

$$P = \bar{\psi}i\gamma_5\psi, \quad (64)$$

where  $\bar{\psi} = \psi^\dagger\gamma^0$  is the Dirac adjoint and  $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$ .

### 7.2 Transformation properties

The five bilinears transform under the Lorentz group  $SO(3,1)$  as:

- $S$ : scalar.
- $V^\mu$ : vector.
- $T^{\mu\nu}$ : antisymmetric rank-2 tensor.
- $A^\mu$ : axial vector.
- $P$ : pseudoscalar.

### 7.3 Construction from the substrate

The bilinears are constructed from the Hodge fiber by forming the Dirac adjoint  $\bar{\psi} = \psi^\dagger\gamma^0$  and sandwiching the appropriate gamma matrix structure. The gamma matrices  $\gamma^\mu$  are given by Theorem 3.1. The chirality operator  $\gamma_5$  is given by (38).

**Proposition 7.2** (Fierz completeness). *The five Dirac bilinears form a complete basis for  $4 \times 4$  matrices on the Hodge fiber. Any bilinear  $\bar{\psi}_1\Gamma\psi_2$  can be expanded in the five basis structures.*

*Proof.* The Clifford algebra  $\text{Cl}(3,1)$  has dimension  $2^4 = 16$  as a real vector space. A basis is provided by the 16 elements

$$\{\mathbb{I}_4, \gamma^\mu, \sigma^{\mu\nu}, \gamma^\mu\gamma_5, i\gamma_5\}, \quad (65)$$

with dimensions  $1 + 4 + 6 + 4 + 1 = 16$ . This is the standard basis of the Clifford algebra and spans the space of  $4 \times 4$  matrices.  $\square$

## 8 Lattice Corrections

### 8.1 Exact lattice dispersion relation

The exact dispersion relation for the Hodge-Dirac operator on  $\Lambda$  is

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (66)$$

For a massive mode with mass  $m$ , the exact lattice dispersion is

$$\omega^2(n_1, n_2) = \lambda(n_1, n_2) + m^2. \quad (67)$$

### 8.2 Expansion at low momentum

Using the identity  $4 - 2 \cos \theta = 4 \sin^2(\theta/2)$ ,

$$\lambda(n_1, n_2) = 4 \sin^2\left(\frac{\pi n_1}{8}\right) + 4 \sin^2\left(\frac{\pi n_2}{8}\right). \quad (68)$$

In terms of physical momentum  $p_i = \pi n_i/(4a)$ ,

$$\frac{\lambda}{a^2} = 4 \sin^2\left(\frac{ap_1}{2}\right) \frac{1}{a^2} + 4 \sin^2\left(\frac{ap_2}{2}\right) \frac{1}{a^2}. \quad (69)$$

Expanding in  $a$ ,

$$\frac{\lambda}{a^2} = \mathbf{p}^2 - \frac{a^2}{12}(p_1^4 + p_2^4) + \frac{a^4}{360}(p_1^6 + p_2^6) + O(a^6). \quad (70)$$

### 8.3 $O(\epsilon^2)$ correction to the Dirac dispersion

**Theorem 8.1** (Lattice-corrected Dirac dispersion). *The lattice-corrected Dirac dispersion relation is*

$$\omega^2 = \mathbf{p}^2 + m^2 - \frac{a^2}{12}(p_1^4 + p_2^4) + O(a^4). \quad (71)$$

Equivalently, in terms of the lattice-to-observation scale ratio  $\epsilon = a/\ell_{\text{obs}}$ ,

$$\omega^2 = \mathbf{p}^2 + m^2 \left[ 1 + \frac{1}{18}\epsilon^2 + O(\epsilon^4) \right]. \quad (72)$$

*Proof.* The exact dispersion is  $\omega^2 = \lambda(n_1, n_2) + m^2$ . Substituting the expansion of  $\lambda/a^2$  gives the first form. The second form follows by expressing  $ap_i$  in terms of  $\epsilon$  and normalizing the mass term.  $\square$

### 8.4 Suppression at observable scales

At all observable scales,  $\epsilon^2 \lesssim 10^{-70}$ , so the lattice correction to the Dirac dispersion relation is unobservable. The continuum Dirac equation is recovered to high precision.

## 9 The Spinor Representation of $SO(3, 1)$ from $\ker D$

### 9.1 The vector representation on $\ker D$

The kernel of the Hodge-Dirac operator satisfies  $\ker D \cong \mathbb{R}^{3,1}$ , carrying the vector representation of  $SO(3, 1)$  via the Hodge star and intersection form (HCSM-03).

## 9.2 The spinor representation from the Hodge fiber

**Theorem 9.1** (Spinor representation). *The four-component Hodge fiber  $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$  carries the spinor representation of  $SO(3,1)$  under the Clifford algebra action generated by the substrate operators  $A_1, A_2$ .*

*Proof.* The Clifford algebra  $\text{Cl}(3,1)$  has a unique irreducible representation of complex dimension 4, the Dirac spinor representation. By Theorem 3.1, the four-component Hodge fiber carries this representation via the substrate generators  $A_0, A_1, A_2, A_3$ . The  $SO(3,1)$  action is generated by commutators  $[\gamma^\mu, \gamma^\nu]$ , which act as Lorentz transformations on the spinor indices.  $\square$

**Corollary 9.2** (Tensor decomposition). *The tensor product  $V_{56} \cong V_{14} \otimes \ker D$  carries the representation  $V_{14} \otimes \mathbf{4}$  of  $SO(3,1)$ , where  $V_{14}$  transforms under the internal  $D_4$  symmetry and  $\mathbf{4}$  is the Dirac spinor representation of  $SO(3,1)$ .*

## 9.3 Relationship between the 4-component fiber and physical spinor

The four-component Hodge fiber at each site is identified with a Dirac spinor of  $SO(3,1)$ . The identification is as follows:

- The  $\Omega^0$  component is the left-handed Weyl spinor  $\psi_L$  (up to basis).
- The  $\Omega^2$  component is the right-handed Weyl spinor  $\psi_R$  (up to basis).
- The two components of  $\Omega^1$  complete the Dirac structure.

The Weyl representation is the natural basis for the Hodge fiber.

# 10 Zero Modes and Antiparticles

## 10.1 Zero modes of $D$

The kernel of the Hodge-Dirac operator is

$$\ker D = H^0(T^2) \oplus H^1(T^2) \oplus H^2(T^2), \quad (73)$$

with dimension  $1 + 2 + 1 = 4$ .

## 10.2 Positive and negative energy solutions

The Fourier symbol  $M(p)$  has eigenvalues  $\pm i|p|$ . The positive eigenvalue corresponds to particle solutions, and the negative eigenvalue to antiparticle solutions. This is the standard Dirac structure.

**Proposition 10.1** (Particle-antiparticle structure). *The Hodge-Dirac operator on the four-component fiber admits both positive and negative energy solutions, corresponding to particles and antiparticles.*

*Proof.* The eigenvalues of  $M(p)$  are  $\pm i|a|$ , with  $|a|^2 = \lambda(n_1, n_2) \geq 0$ . Each eigenvalue has multiplicity 2. The positive eigenvalue subspace is the particle subspace, and the negative eigenvalue subspace is the antiparticle subspace. The charge conjugation operator  $C = i\gamma^2\gamma^0$  maps one to the other.  $\square$

### 10.3 Charge conjugation

**Definition 10.2** (Charge conjugation). The charge conjugation operator is

$$C = i\gamma^2\gamma^0. \quad (74)$$

The charge conjugation operator satisfies

$$C\gamma^\mu C^{-1} = -(\gamma^\mu)^T, \quad C^2 = -\mathbb{I}_4. \quad (75)$$

### 10.4 Majorana condition

**Definition 10.3** (Majorana spinor). A Majorana spinor satisfies

$$\psi = C\bar{\psi}^T. \quad (76)$$

For the Hodge fiber, the Majorana condition is compatible with the substrate structure: the components of  $\psi$  can be chosen real, and the charge conjugation operator maps the substrate's real Hodge fiber to itself.

**Proposition 10.4** (Majorana compatibility). *The Hodge fiber admits a real (Majorana) structure compatible with the Clifford algebra.*

*Proof.* The substrate operators  $A_1, A_2, A_3, A_0$  are real (up to explicit factors of  $i$  in  $A_0$ ). Their commutators generate a real form of the spinor representation. The Majorana condition selects the real subspace of the spinor space.  $\square$

## 11 Dirac vs. Weyl vs. Majorana

### 11.1 Chirality decomposition

Under the chiral projectors  $P_L, P_R$ , a Dirac spinor decomposes as

$$\psi = \psi_L + \psi_R, \quad \psi_L = P_L\psi, \quad \psi_R = P_R\psi. \quad (77)$$

Each of  $\psi_L, \psi_R$  is a two-component Weyl spinor.

### 11.2 Dirac spinor

A Dirac spinor carries both chiralities, related by the mass term. The Dirac equation is

$$(i\gamma^\mu\partial_\mu + m)\psi = 0. \quad (78)$$

In chiral components,

$$i\sigma^\mu\partial_\mu\psi_L = m\psi_R, \quad (79)$$

$$i\bar{\sigma}^\mu\partial_\mu\psi_R = m\psi_L. \quad (80)$$

### 11.3 Weyl spinor

A Weyl spinor carries only one chirality. The Weyl equation is

$$i\sigma^\mu\partial_\mu\psi_L = 0 \quad (81)$$

for a left-handed Weyl spinor, or

$$i\bar{\sigma}^\mu\partial_\mu\psi_R = 0 \quad (82)$$

for a right-handed Weyl spinor.

## 11.4 Majorana spinor

A Majorana spinor is a Dirac spinor satisfying the reality condition  $\psi = C\bar{\psi}^T$ . It carries both chiralities but has only two real degrees of freedom instead of four.

## 11.5 Character of the HCSM spinor

**Theorem 11.1** (Character of the HCSM spinor). *The HCSM spinor at each site is a Dirac spinor of  $SO(3,1)$ . It admits a Majorana reduction when the mass term vanishes. It decomposes into Weyl components under the chiral projectors.*

*Proof.* The four-component Hodge fiber carries the Dirac spinor representation of  $Cl(3,1)$  by Theorem 3.1. The chiral projectors  $P_L, P_R$  decompose it into two Weyl components. The Majorana reduction is available when the mass term vanishes; for massive modes, the Dirac structure is necessary to accommodate the mass term.  $\square$

# 12 Lorentz Covariance of the Lattice Dirac Equation

## 12.1 Exact Lorentz covariance in the continuum limit

The continuum limit of the Hodge-Dirac operator is the standard Dirac operator  $i\gamma^\mu\partial_\mu$ , which is exactly Lorentz covariant. Any deviation from Lorentz covariance is  $O(\epsilon^2)$  in the lattice-to-observation scale ratio.

## 12.2 Lattice artifacts

At finite lattice spacing, the discrete Dirac operator breaks exact Lorentz covariance. The lattice dispersion relation is

$$\omega^2 = \mathbf{p}^2 + m^2 - \frac{a^2}{12}(p_1^4 + p_2^4) + O(a^4), \quad (83)$$

which is not exactly Lorentz covariant. The violation is  $O(a^2)$  and is suppressed by  $\epsilon^2$  at observable scales.

**Theorem 12.1** (Lorentz covariance). *The HCSM Dirac equation is exactly Lorentz covariant in the continuum limit  $\epsilon \rightarrow 0$ , and approximately Lorentz covariant at finite  $\epsilon$  with corrections of order  $\epsilon^2$ .*

# 13 Summary of Results

The Dirac equation from the Hodge-Dirac operator is fully characterized by the following results.

1. The Hodge fiber  $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$  has real dimension 4 (Section 6).
2. The Fourier symbol  $M(p)$  of the Hodge-Dirac operator satisfies  $M(p)^2 = |p|^2 \mathbb{I}_4$  (Theorem 2.4).
3. The two substrate momentum operators  $A_1, A_2$  generate a Clifford algebra  $Cl(3,1)$  with the additional generators  $A_3 = A_1 A_2$  and  $A_0 = iA_3$  (Theorem 3.1).
4. The chirality operator  $\gamma_5 = -iA_1 A_2$  satisfies  $\gamma_5^2 = \mathbb{I}_4$  and  $\{\gamma_5, \gamma^\mu\} = 0$  (Theorem 4.2).
5. The chirality operator  $\gamma_5$  is unitarily equivalent to the Hodge form-parity  $(-1)^F$  (Theorem 4.7).
6. An explicit unitary  $U$  transforms  $M(p)$  into the standard Dirac form  $i\gamma^\mu p_\mu$  (Theorem 5.1).

7. The continuum limit of the Hodge-Dirac operator is the Dirac operator  $i\gamma^\mu\partial_\mu$  (Theorem 5.2).
8. The mass term in the Dirac equation is the HCSM mass map  $m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)i)$  (Definition 6.1).
9. The mode-dependent Dirac equation on the 56-dimensional sector is  $(D_{56} + m_i)\psi_i = 0$  (Theorem 6.2).
10. The Dirac bilinears form a complete basis for  $4 \times 4$  matrices on the Hodge fiber (Proposition in Section 7).
11. The lattice corrections to the Dirac dispersion relation are  $O(\epsilon^2)$  (Theorem 8.1).
12. The Hodge fiber carries the Dirac spinor representation of  $SO(3,1)$  (Theorem in Section 9).
13. The Dirac equation admits both particle and antiparticle solutions (Proposition in Section 10).
14. The HCSM spinor admits a Majorana reduction when the mass term vanishes (Theorem in Section 11).
15. The HCSM Dirac equation is exactly Lorentz covariant in the continuum limit (Theorem in Section 12).

## 14 Conclusion

We have derived the Dirac equation from the Hodge-Dirac operator on the periodic torus  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ . The four-component Hodge fiber at each site carries the Dirac spinor representation of the Clifford algebra  $\text{Cl}(3,1)$ , generated by the two native substrate momentum operators together with two additional generators constructed from their product. The chirality operator  $\gamma_5$  is the substrate-native chirality grading, unitarily equivalent to the Hodge form-parity  $(-1)^F$ . The continuum limit of the Hodge-Dirac operator is the standard Dirac operator  $i\gamma^\mu\partial_\mu$ , up to an explicit unitary transformation. The mass term is the HCSM mass map, giving a mode-dependent Dirac equation on the 56-dimensional Hodge sector. The lattice corrections are  $O(\epsilon^2)$  and unobservable at all accessible scales. The Dirac bilinears, chiral projectors, particle-antiparticle structure, and Majorana reduction are all constructed from the substrate. The HCSM Dirac equation is exactly Lorentz covariant in the continuum limit.

## A Explicit Construction of the Unitary $V$

The unitary  $V$  relating  $\gamma_5$  and  $(-1)^F$  is constructed explicitly. In the basis where  $(-1)^F = \text{diag}(+1, -1, -1, +1)$ , choose

$$V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & i & 0 \\ 0 & 1 & 0 & -i \\ -i & 0 & 1 & 0 \\ 0 & i & 0 & 1 \end{pmatrix}. \quad (84)$$

Direct computation gives  $V^\dagger(-1)^F V = \gamma_5$ .

## B Explicit Computation of the Continuum Limit

The Fourier symbol  $M(p)$  at small momentum is

$$M(p) = p_1 A_1 + p_2 A_2 + O(p^2). \quad (85)$$

The unitary  $U$  in (49) satisfies

$$U^\dagger A_1 U = \gamma^1, \quad U^\dagger A_2 U = \gamma^2, \quad U^\dagger A_3 U = \gamma^3, \quad U^\dagger A_0 U = \gamma^0, \quad (86)$$

where the gamma matrices are in the Weyl representation. Therefore

$$U^\dagger M(p) U = i(p_1 \gamma^1 + p_2 \gamma^2 + p_3 \gamma^3 + p_0 \gamma^0) + O(p^2), \quad (87)$$

with  $p_3 = 0$  on the substrate and  $p_0 = |p|$  for massless modes.

## C Clifford Algebra Details

The four generators  $A_0, A_1, A_2, A_3$  in the fiber basis  $(\Omega^0, \Omega_1^1, \Omega_2^1, \Omega^2)$  are

$$A_0 = \begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & -i & 0 \end{pmatrix}, \quad (88)$$

$$A_1 = \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & -i \\ i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix}, \quad (89)$$

$$A_2 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}, \quad (90)$$

$$A_3 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}. \quad (91)$$

Direct computation verifies  $\{A_\mu, A_\nu\} = 2\eta_{\mu\nu} \mathbb{I}_4$  with  $\eta = \text{diag}(+1, +1, +1, -1)$ .

## D Numerical Verification

The results of this paper have been verified at machine precision in IEEE-754 double precision. The verification script `HCSM-65_verification.py` checks:

1. The Fourier symbol  $M(n_1, n_2)$  in (15).
2. The square identity  $M(p)^2 = |p|^2 \mathbb{I}_4$  for all momenta.
3. The Clifford algebra  $\{A_\mu, A_\nu\} = 2\eta_{\mu\nu} \mathbb{I}_4$ .
4. The chirality operator  $\gamma_5 = -iA_1A_2$ .
5. The unitary equivalence  $\gamma_5 = V^\dagger(-1)^F V$ .

6. The explicit unitary  $U$  in (49).
7. The continuum limit  $U^\dagger M(p)U = i\gamma^\mu p_\mu$ .
8. The mode-dependent Dirac equation on the 56-dimensional sector.
9. The lattice corrections through  $O(\epsilon^2)$ .

## E Glossary

**Hodge-Dirac operator.**  $D = d + d^*$ , acting on the Hodge complex.

**Kähler-Dirac operator.** The mathematical name for the operator  $d + d^*$  on differential forms.

**Hodge fiber.** The four-component fiber  $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$  at each site.

**Chirality operator.**  $\gamma_5 = -iA_1A_2$ , the substrate-native chirality grading.

**Hodge form-parity.**  $(-1)^F$ , the  $\mathbb{Z}_2$  grading of the Hodge complex.

**Mass map.**  $m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i)$ , the HCSM mass assignment.

**Lattice-to-observation scale ratio.**  $\epsilon = a/\ell_{\text{obs}}$ , the ratio of the structural lattice spacing to the observation scale.

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